12 Point **Q** is a point between two planets of different masses where an object will experience zero resultant gravitational force due to the two planets.

Which statement about point **Q** is correct?

A A mass moved along the line from point **Q** to the surface of the planet of smaller mass experiences an increase in gravitational potential energy.

B The total gravitational potential at point **Q** is zero.

C The gravitational potential at point **Q** depends on the mass of any object placed there.

D The gravitational potential gradient at point **Q** is zero.